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Lab 5 part A and B

Industrial Control Devices: Switches, Indicators, relays, solenoids, and contactors

Introduction

Chapter 2 of the book highlights the wide range of components commonly used in industrial systems. Even seemingly simple devices like switches come in various types with different applications. In future labs, we will explore several of these components, including pneumatic solenoids, timers, counters, and control transformers. Additionally, electromechanical relays and solid-state relays (SSRs) will be examined in greater detail in an upcoming optocoupler lab.

What is a Relay?

A relay is an electrically operated switch that uses an electromagnet to control high-power circuits with a low-power signal.

Uses of a Relay:

- Industrial automation
- Automotive systems
- Power distribution
- Microcontroller circuits

Key Relay Specifications:

- Coil Voltage & Current (e.g., 5V, 12V, 24V DC/AC)
- Contact Ratings (max voltage/current it can switch)
- Internal Configuration: SPST, SPDT, DPDT
- Pin Layout: COM, NO, NC, Coil Pins

What is a Solenoid?

A solenoid converts electrical energy into mechanical motion using a coil and a movable plunger.

Uses of a Solenoid:

- Electromechanical locks
- Automotive starters
- Pneumatic & hydraulic valves
- Industrial actuators
- Key Solenoid Specifications:
- Operating Voltage & Current (e.g., 12V, 24V DC)
- Stroke Length (distance plunger moves)
- Holding & Inrush Current (initial high current needed to activate)

Part A

Step 1: The following switches listed in the lab handout were reviewed, and items 1-7 were determined for each switch based on the data sheets provided in the lab.

Switch Model	Configuration	Operating Voltage	Current Rating	Operating lifetime	Environmental conditions (IPXY rating)	Size and mounting	Other options (lights, key operations)
C&K 7301	3PDT	120V AC 28V DC	5A AC 2A DC	100,000 cycles	IP67	15/32-32 Threaded	Satin finish
APEM 5237 2-F 2	DPDT	250V AC 30V DC	5A AC	100,000 cycles	IP67	Panel mount	Sealing options
McGill 2603-1150	SPDT DPDT	125 AC 250 AC	20 A	100,000 cycles	IP65	27.7mm x16.5mm x6.9 mm Screw	Actuator Thick terminals
Honeywell Microswitch V3I-139-D8	SPDT SPST	250V DC	15.1 A	100,000 cycles	IP65	27.7mm x16.5mm x6.9 mm Screw	Actuator Thick terminals
EAO 51 - 255-025K	SPDT SPST	250V AC	6A	1M cycles	IP65	24mm x 24mm Panel Mount	Ilumination Colors
E1A50 1NC1NO	NC/NO	250 V AC 24 V DC	3A to 10A	100K to 500K	IP65	22mm or 30 mm Diameter Panel	Ilumination Key reset

Table 1. *Various switches and their data.*

Step 2: Indicators: c3controls 13SBLG24ST, or APT AD16-22D/S23 are defined and evaluated for information about items 1-7.

- The c3controls 13SBLG24ST-13GNR is a 13mm pilot green light indicator designed for industrial applications.

- The APT AD16-22D/S23 is a 22mm LED indicator light commonly used in industrial applications

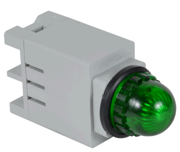
Image	Configuration	Operating Voltage	Current Rating	Operating Lifetime	Size and Mounting	Enviromental Conditions
	c3controls 13SBLG24ST	24 V DC 24 V AC	5mA to 20 mA	50,000 hours	13 mm diameter Screw terminal	4X waterlight applications
	APT AD16-22D/S23	24 V DC 24 V AC 220 V AC 220 V DC	5mA to 20mA	30,000 hours	22mm diameter Panel mount	IP54 IP 65 IP 67

Table 2. *Information about indicators c3controls and APT AD16-22D/S23.*

Step 3: Rheostats, i.e. Power Variable Resistors or Power Pots are defined and information concerning items 1-7 are found.

- A rheostat is a variable resistor that controls current in a circuit by manually adjusting resistance. It is commonly used in applications where the amount of electrical resistance needs to be varied, such as light dimmers, motor speed controls, and heating elements.

Key Features of a Rheostat:

- Two-Terminal Connection: Unlike potentiometers (with three terminals), rheostats typically use only two terminals—one connected to a fixed end of the resistive element and the other to a sliding contact (wiper).
- Resistive Element: It's made of materials like wire-wound resistors (nichrome wire) or carbon composition.
- Manual Adjustment: The resistance changes when a sliding or rotating contact is moved along the resistive element.
- High Power Handling: Rheostats are designed to handle high current and power loads, making them useful in industrial and laboratory applications.

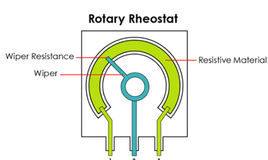
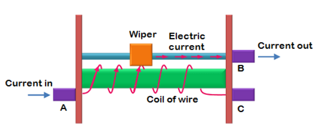
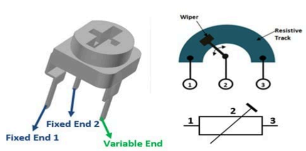
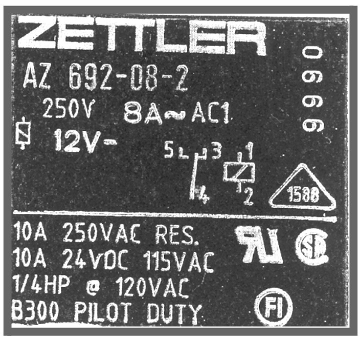
	● Rotary Rheostat: Uses a circular resistive element with a rotating knob to adjust resistance.
	● Linear Rheostat (Slider Rheostat): A sliding wiper changes resistance along a straight resistive track.
	● Preset Rheostat: It is a small, factory-set rheostat used for circuit calibration.

Table 3. *Types of Rheostats*

Applications of Rheostats:

- Motor Speed Control
- Light Dimmers
- Heater Control
- Voltage and Current Regulation

Part B



IMG 1. *The relay, analyzed in table 4 on the next page.*

Step 1-3: analyzing the relay

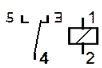
Item	12	Maximum voltage for the input for the relay
	-8	Maximum input current for the relay
1	Manufactor	Zettler
2	Model	AZ 692-08-2
3	Date code	99990
4	B300 pilot duty	Relay is designed to handle inductive loads
5	250V 8A ~AC1	Maximum switching current is 8A, and maximum switching voltage is 250V.
6	10A 250VAC RES.	RES means resistive load at 250V AC and 10 A current
7	10A 24VDC 115VAC	This relay can be used with 115 AC on 24V DC voltage for 10A switching current
8	1/4 HP @ 120VAC	Maximum switch power is 1.4 hp. Recall one HP = 746 watts and 120 vac is the switch
9		Pin layout
10	12V-	Coil operation 12 V
11	SA 1588 FI	Certifications

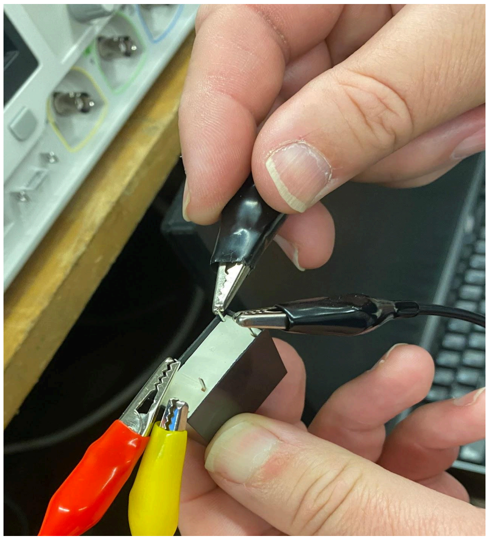
Table 4. *Information about the relay used in the lab. The link to the data sheet for this relay is here:* [*https://www.digchip.com/datasheets/parts/datasheet/540/AZ692-08-2-pdf.php*](https://www.digchip.com/datasheets/parts/datasheet/540/AZ692-08-2-pdf.php)

Step 4: creating a test circuit to test the relay

The information needed for a test circuit is:

- The minimum coil voltage required for operation, starting with 6 V.
- Voltage type
- Voltage limit (12V)- we must be careful to not exceed this limit.

To determine the voltage required for a relay, always refer to the relay's datasheet or markings for the specified "coil voltage," which indicates the voltage necessary to energize the relay coil and activate the contacts.



IMG 2. *The test circuit*

First we started with mid-range voltage and increased the voltage because we couldn't hear the clicking sound from the relay. One we heard the clicking sound, we decreased the voltage until we couldn't hear it anymore. When using the test circuit, we made sure to not exeed 12V. As a result, we found that at 7 volts it switched on and we heard a clicking sound, and at lower voltage we got an open circuit at about 3 volts and it drops out at 3.7 miliamps.

Step 5:

Once part 4 was complete, we started at 0v for the coil voltage and increased the voltage, and monitored the current until the relay activates or pulls in until the input voltage is increased to the coils nominal (rated) voltage. From the rated voltage, we reduced the voltage until the relay "drops out". We checked these values in the data sheet to double check and it matched the value in table 4. If we plotted the relay condition (off-0 or on-1) verses the input voltage from 0v to the relay coil voltage in steps of 1v, the plot would display a difference between the pull-in and drop out voltages, like a logic chip.

Next, we reversed the coil voltage and repeated the process to determine if the relay would still operate under reversed polarity. We recorded the pull-in and drop-out voltages under these conditions and noted any differences in coil current before and after activation. This helped us understand the relay's response to voltage changes and its polarity sensitivity.The results confirmed table 4.

Step 6:

The data sheet for the DPDT relay was reviewed, as it will be used in the next lab. The DPDT Panasonic TQ2-24V data sheet that was reviewed can be found here: https://www.mouser.com/datasheet/2/315/mech_eng_tq-1299280.pdf. We paid special attention to the coil pins and DPDT action, and it's switched contacts/ connections marked as COM/NO/NC.

Step 7: Solenoid operation.

Solenoids are similar to relays, but the physical movement is the desired output instead of the contact switching as the desired output. Typical applications include home or automotive door locks.

We did a google search for "Universal Car Power Door Lock Actuator". With this unit, the output is the "force" generated by the system, so a "spring" or "fish-type" scale will be used. Rather than lift a weight to measure the force, we will simply restrain or "hold" the scale so when activated, the force can be read.

We found that a Universal Car Power Door Lock Actuator is a small electric motor mechanism installed inside a car door to control the locking and unlocking of the door automatically. It is commonly used in vehicles that do not have factory-installed power locks or as a replacement for a faulty actuator.

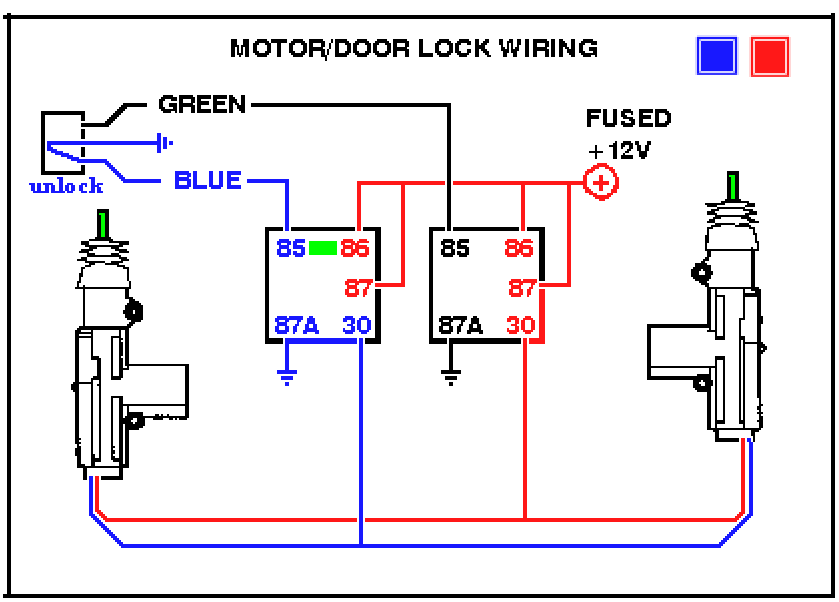
How It Works:

- The actuator comprises a small electric motor, gears, and a rod connecting to the car's locking mechanism. When you press the lock or unlock button on a remote key fob, switch, or alarm system, the actuator receives an electric signal to operate the locking mechanism.

- This signal makes the motor move a rod or lever that either pushes or pulls the lock mechanism, locking or unlocking the door.

- The actuator can be installed in both manual and automatic locking systems.

It is called universal because it is designed to fit most car models, even those that didn't originally come with power locks. It typically comes with a simple mounting bracket and can be adapted to various types of doors.



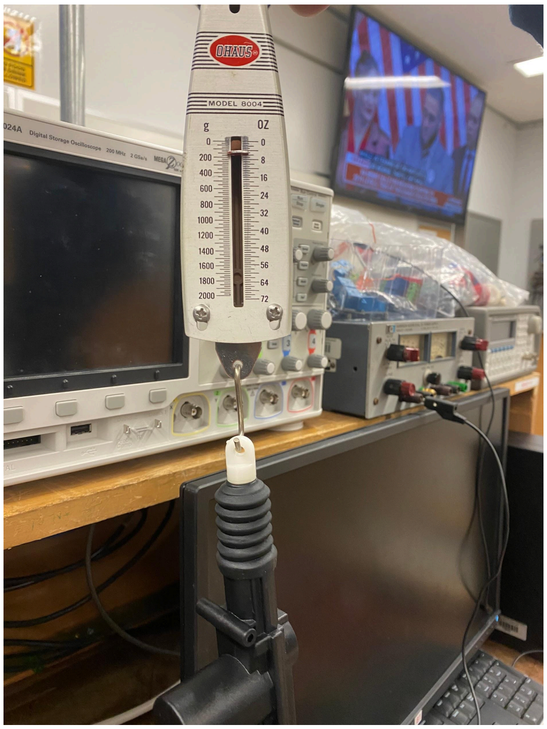
Common Uses:

- With only manual locks, power locks should be added to older or base-model vehicles.

- Replacing a defective actuator in an existing power lock system.

- Integrating with a keyless entry or alarm system.

IMG 3. *Universal Car Power Door Lock Actuator*



IMG 4. *Solenoid system*

The solenoid is a push pull, 12v system with 400g.

Step 8. Examination of a contactor, the Tyco P41C47DHO103 contactor operation

The Tyco P41C47DHO103 is a specific model in the P41 series of contactors, produced by TE Connectivity's Potter & Brumfield Relays division. Contactors are electrically controlled switches used to manage electrical power circuits. They operate similarly to relays but are designed to handle higher current ratings.

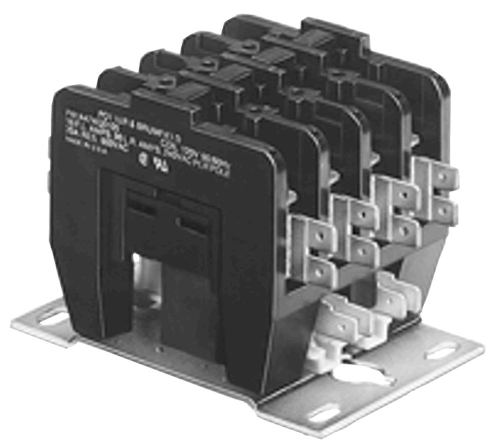
Key Features of the P41 Series Contactors:

- The P41 series provides various pole configurations, including options for 3-pole and 4-pole variants.
- Coil Options: Offered with both AC and DC coils, allowing flexibility for various control circuit requirements.
- Optional auxiliary switches, such as signaling or interlocking, can be added for enhanced functionality.
- A range of main contact terminals is available to meet various wiring requirements.
- Applications: Commonly used in HVAC systems and industrial control applications.

Operation:

When the correct voltage is applied to the contactor's coil, it creates a magnetic field that pulls the contacts together, closing the circuit and allowing current to flow to the connected load. When the coil is de-energized, the magnetic field dissipates, and the contacts return to their open position, interrupting the current flow.

P31



P41

The voltage and current for operation of the P31/P41 series Definite Purpose Magnetic Contactor are as follows:

Coil Voltage:	Nominal Coil Current:	Must-Operate Voltage:
12V DC (Code: FO) 24V DC (Code: HO)	For 12V DC: 571 mA For 24V DC: 286 mA	For 12V DC: 9V For 24V DC: 18V

Table 5. *Voltage and current of the P31/P41 series Definite Purpose Magnetic Contactor. These values are specified in the Coil Data table provided in the datasheet.*

Conclusion:

In this lab, we explored the fundamental components used in industrial systems, focusing on relays, solenoids, and other control devices. We examined how relays function and the importance of understanding their operating voltage ranges, and how to gather necessary information from data sheets, creating/ utilizing test circuits, and interpreting markings on the component itself. Additionally we studied solenoids and their applications, as well as specific examples such as universal car power door lock actuators and contactors. This hands-on experience reinforced our understanding of these essential components, and also had the benefit of preparing us for future labs that will utilize these components and build on these concepts.